

Veda Internship 2026 — Day 01

Command-Line Calculator (Calculator.java)

A Java console calculator supporting addition, subtraction, multiplication, division and modulus, with input validation and zero-division handling.

Source Code

```
1  import java.util.Scanner;
2
3  public class Calculator {
4
5      // Addition
6      public static double add(double a, double b) {
7          return a + b;
8      }
9
10     // Subtraction
11     public static double subtract(double a, double b) {
12         return a - b;
13     }
14
15     // Multiplication
16     public static double multiply(double a, double b) {
17         return a * b;
18     }
19
20     // Division
21     public static double divide(double a, double b) {
22         return a / b;
23     }
24
25     // Modulus
26     public static double modulus(double a, double b) {
27         return a % b;
28     }
29
30     public static void main(String[] args) {
31
32         Scanner scanner = new Scanner(System.in);
33
34         System.out.println("=====");
35         System.out.println("      COMMAND-LINE CALCULATOR");
36         System.out.println("=====");
37
38         // Read first number
39         double num1;
40
41         while (true) {
42             System.out.print("Enter first number: ");
43
44             if (scanner.hasNextDouble()) {
45                 num1 = scanner.nextDouble();
46                 break;
47             } else {
48                 System.out.println("Invalid input! Please enter a valid number.");
49                 scanner.next();
50             }
51         }
52
53         // Read operator
54         System.out.print("Enter operation (+, -, *, /, %): ");
55         char operator = scanner.next().charAt(0);
56     }
```

```

57 // Read second number
58 double num2;
59
60 while (true) {
61     System.out.print("Enter second number: ");
62
63     if (scanner.hasNextDouble()) {
64         num2 = scanner.nextDouble();
65         break;
66     } else {
67         System.out.println("Invalid input! Please enter a valid number.");
68         scanner.next();
69     }
70 }
71
72 double result;
73
74 // Perform operation
75 switch (operator) {
76
77     case '+':
78         result = add(num1, num2);
79         System.out.println("Result = " + result);
80         break;
81
82     case '-':
83         result = subtract(num1, num2);
84         System.out.println("Result = " + result);
85         break;
86
87     case '*':
88         result = multiply(num1, num2);
89         System.out.println("Result = " + result);
90         break;
91
92     case '/':
93         if (num2 == 0) {
94             System.out.println("Error: Division by zero is not allowed.");
95         } else {
96             result = divide(num1, num2);
97             System.out.println("Result = " + result);
98         }
99         break;
100
101     case '%':
102         if (num2 == 0) {
103             System.out.println("Error: Modulus by zero is not allowed.");
104         } else {
105             result = modulus(num1, num2);
106             System.out.println("Result = " + result);
107         }
108         break;
109
110     default:
111         System.out.println("Invalid operation!");
112         System.out.println("Please use +, -, *, /, or %.");
113 }
114
115 scanner.close();
116 }
117 }

```